



 PAULYNN OPURUM
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 BLOG

Our Green Energy HVAC Future

As we venture into a green energy HVAC future, we begin to find new ways of powering our homes, businesses, devices and lives.

HVAC systems remain a necessity for nearly every building. And a steady supply of energy is crucial to heat, cool and ventilate spaces effectively.

The majority of energy produced in the United States comes from extracting and burning fossil fuels. However, there's been growth in renewable alternative energy technologies.

This field of energy production has been nicknamed *Green Energy*.

For the HVAC industry, green energy generated by collecting solar and geothermal energy can be used to heat and cool residential structures.

What Is Green Energy?

In simple terms, renewable energy is produced from exhaustible resources. The Earth's supply of fossil fuels like oil and coal will eventually run out. However, the sun will continue to provide free energy to this planet for billions of years.

Green Energy HVAC Systems

Solar Energy

As its name suggests, solar energy is generated from capturing sunlight.

Photovoltaic cells absorb the sun's rays and convert them into electricity through solar panels. The energy is captured in batteries or sold back to the grid (where allowed) and can supply power to a building's HVAC system.

Pros: Most of the Earth's population has access to sunlight, making solar energy a widely accessible renewable source for HVAC.

Cons: Solar panels can save more money in the long run; however, they are expensive to purchase and install. Also, complex and costly battery arrays are necessary to collect enough energy to run HVAC systems overnight.

Geothermal Energy

Geothermal energy is heat that radiates from the rocks and fluids underneath the Earth's crust. By digging tunnels and creating wells in the ground, steam and hot water can be used to drive turbines connected to electricity generators.

A **geothermal system** consists of an indoor handling unit and pipes that are buried in the ground. These pipes (earth loops) connect from a pump to a reinjection well that provides steam. Since the temperature in the Earth's crust is constant, it provides free energy.

In winter, the fluid that circulates through the system's earth loop is carried indoors. The indoor unit then transforms this heat to a higher temperature and releases it into the building, acting as a heater.

In summer, the geothermal system absorbs heat from the building and transfers it through the earth loop/pump to reinjection well.

It's important to note that geothermal systems do not need as frequent maintenance as most HVAC systems might. Once installed, the earth loop can last for generations. However, filter changes and coil cleaning can help improve system efficiency and longevity.

Pros: Geothermal energy is essentially self-replenishing, and has little impact on surface ecosystems.

Cons: Geothermal systems are expensive to build, and certain areas of the world that are prone to earthquakes would not be suitable for geothermal systems.

When planning a green energy HVAC system for a customer, you'll need to adjust your pricing model to account for the different hardware necessary. Download our FREE HVAC Pricing Calculator today!

Download Now

Our Future Is Green Energy HVAC

As HVAC technologies advance, we begin to find new ways to reduce the negative environmental impacts created by burning fossil fuels for energy generation.

Green energy sources like solar and geothermal energy can produce reliable and renewable energy to power our homes and businesses. As we begin to use renewable technologies, we will see improvements to our environment and increased efficiency in green energy HVAC systems.

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